Forest Management Plan

Land Sciences

May, 2026

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Forest Management Plan (A11664)

Short Title: Forest Management Plan
Faculty Science and Computing
Department Land Sciences
Cluster Forestry
Author MPEDINI
Credits 10 **Level:** Intermediate

Description of Module / Aims

This is a 10 credit module whose final outcome is a comprehensive five-year Forest Management Plan for a medium-sized property. The demands on forestry as a provider of timber, recreation, environmental and social benefits are ever increasing and complex. In order to reach a sustainable balance, foresters need planning tools. In this module students will experience the planning process of a Forest Management Plan. The focus of the module will allow students to gain professional skills in using models for timber growth, and calculating clear-fell and thinning yield. In preparing a Forest Management Plan for a five-year period the student will have to synthesise all the knowledge and skills acquired over the three programme years. Preparing a plan will have to be done with the forest owner's aims and objectives in mind, as well as the constraints given by legislation, environmental guidelines and local social issues.

Programmes

`PROJ-0005`	BSc in Forestry (`WD_SFORS_D`)	3 / 6 / M

Indicative Content

- Categories of forest management plans
- Planning aims and objectives
- The legal and regulatory framework for forest management plans
- The structure and preparation of a forest management plan
- Forest Management Software/Forest Information Systems
- The yield class/site index system
- Static yield tables and dynamic yield models
- Forecasting timber yield, assortment distribution and future standing volume
- Calculating annual budget for timber revenue, operations costs and forest valuation
- Assessing wind-throw hazard and other growing conditions
- Species dependent silvicultural cost models
- Managing biodiversity in a forestry context
- Analysis and presentation of forest management data
- Evaluating a forest management plan in relation to the owner's objectives, management constraints and sustainability
- Submission of forest management plans through the Forest Service IFORIS/iNET portal

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Utilise the concepts of strategic, tactical, and operational forest planning in the context of a forest management plan.
2. Analyse the aims and constraints given by the forest owner in order to formulate operational management objectives.
3. Utilise the British Forestry Commission's Yield Class system and the internationally used site index.
4. Utilise both static and dynamic yield models as tools for making prognosis of future timber growth, timber yield and harvest revenue.
5. Collect and compile data from different sources and describe a forest comprehensively in terms of its economic, social, and environmental values.
6. Designate prescriptions at stand level and develop local cost models for various tree species based on management objectives and local growing conditions.
7. Produce a map-based Forest Management Plan for a five-year period.
8. Analyse the financial implication of the planning decisions in terms of annual cash flows and changes in forest value.

Learning and Teaching Methods

- Lectures will be used to introduce the theory.
- Field trips will be used to demonstrate field work procedures and for students to carry out field registration, as well as meeting foresters on site to experience how planning decisions are carried through in the management of a forest property.
- Computer practicals will be used to generate the geodatabase for the forest property data, and to prepare maps and cost and revenue forecasts.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	25%	2,3,4
Continuous Assessment	75%	
Assignment	25%	3,4
Project	50%	1,2,5,6,7,8

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/ skill and understanding to carry out assessments. Failure to meet the objectives of assessments. Unable to carry out or submit independent study and research. Unable to present written work with attention to correct formatting, grammar and spelling.

40%-49%: Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on assessments. Attempt made to meet objectives and carry out independent study and research. Final result lacks balance and arguments undeveloped. Can present written work correctly formatted with minor grammar and spelling errors only.

50%-59%: Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out assessments. Satisfactory handling and awareness in carrying out independent study and research. Reporting and projects adequately structured and referenced.

60%-69%: Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design and current knowledge limitations when carrying out assessments. Projects and reporting coherent and soundly structured illustrating wide and in depth reading on subject coupled with the proper use of references.

70%-100%: Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on assessments. Show initiative, individual thought, and enthusiasm in self-study and independent learning.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	60	
Field Trips	36	
Independent Learning	174	

Essential Material

- COFORD. _Irish Dynamic Yield Models for Forest Management User Manual_ by Anonymous, A.. Dublin, Ireland. 2010.
- Forest Service, . _Code of Best Forest Practice - Ireland_. Dublin, Ireland: Forest Service, 2000.
- Forestry Commission, F. _User Manual Forest Yield: A PC-based yield model for forest management in Britain_. Edinburgh: Forestry Commission, 2016.
- Mathews, R.W., T.A.R. Jenkins, E.D. Mackie and D.C. Dick. _Forest Y__ield: A handbook on forest growth and yield tables for British forestry_. Edinburgh: Forestry Commission, 2016.
- McCullagh, A., M. Hawkins, L. Broad and M. Nieuwenhuis. "A comparison of two yield forecasting methods used in Ireland." _Irish Forestry_ 70. (2013): 7-17.

Supplementary Material

- "Forest Management Plans." www.coillte.ie. http://www.coillte.ie/coillteforest/plans/forest_management_plans/
- COFORD. _Forest Management Planning Group Summary Report_ Dublin Ireland. 2015.
- Joyce, P.M. "Development of Yield Tables." _Irish Forestry_ 39, 2. (1982): 65-75.

Requested Resources

- Lecture Room: Loose Seated
- Room Type: Computer Lab